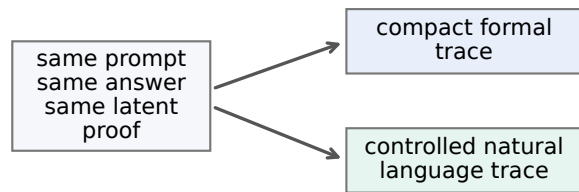


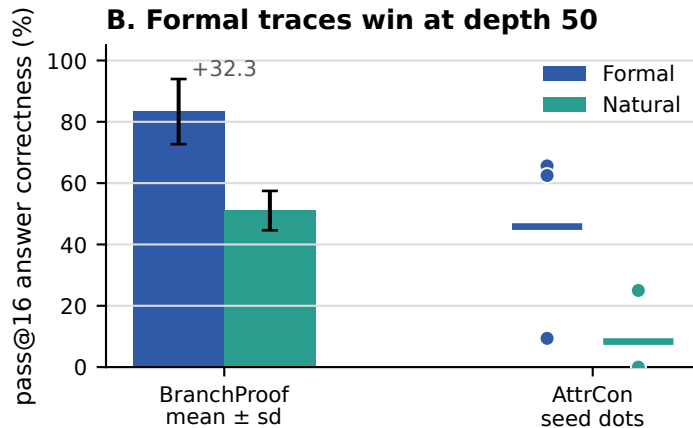
Formal traces improve extrapolative answers, but proof validity remains a separate target

A. Same proof, different trace language

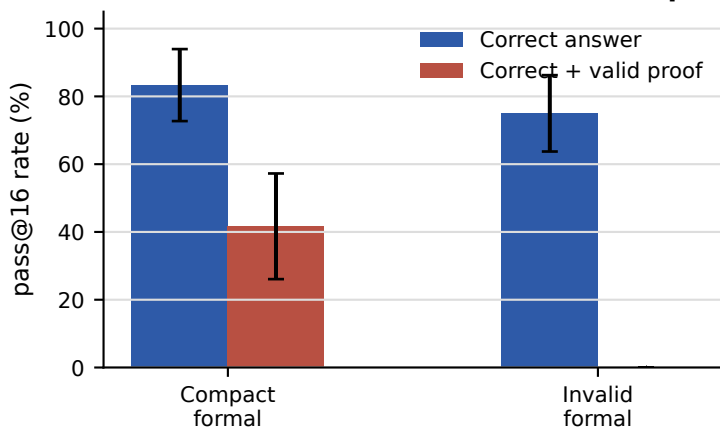


Only the trace substrate changes.

B. Formal traces win at depth 50



C. Correct answers need not be valid proofs



D. Samples expose success and truncation

Formal depth-50 success

```
gold=olive | answer=olive | valid=1
<formal> <constants> a = a b = b c = c
d = d e = e f = f g = g h = h i = i j =
j k = k l = l ...
```

Natural depth-50 failure

```
gold=juniper | answer=missing | valid=0
<think> <premises> a is orchid. a is
east. If a is south and a is orchid,
then b is cedar. I ...
```

Excerpts come from sample_generation_snippets.csv. Metrics use 16 samples per prompt.